

### 2.1.2 Biological molecules

The cells of all living organisms are composed of biological molecules. Proteins, carbohydrates and lipids are three of the key groups of biological

macromolecules that are essential for life. A study of the structure of these macromolecules allows a better understanding of their functions in living organisms.

| Learning outcomes                                                                                                                                               | Additional guidance                                                                                                                                                                                                                                                                                                                     |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                                   | Where appropriate, this section should include diagrams to represent molecular structure and bonding.                                                                                                                                                                                                                                   |
| (a) how hydrogen bonding occurs between water molecules, and relate this, and other properties of water, to the roles of water for living organisms             | To include roles that relate to the properties of water; solvent, transport medium, coolant and as a habitat<br><b>AND</b><br>roles illustrated using examples of prokaryotes and eukaryotes.<br><br>Learners should be able to apply their knowledge and understanding in the context of prokaryotes and eukaryotes.<br><br>HSW2, HSW8 |
| (b) the concept of monomers and polymers and the importance of condensation and hydrolysis reactions in a range of biological molecules                         |                                                                                                                                                                                                                                                                                                                                         |
| (c) the chemical elements that make up biological molecules                                                                                                     | To include:<br><br>C, H and O for carbohydrates<br>C, H and O for lipids<br>C, H, O, N and S for proteins<br>C, H, O, N and P for nucleic acids                                                                                                                                                                                         |
| (d) the ring structure and properties of glucose as an example of a hexose monosaccharide and the structure of ribose as an example of a pentose monosaccharide | To include the structural difference between an $\alpha$ - and a $\beta$ -glucose molecule<br><b>AND</b><br>the difference between a hexose and a pentose monosaccharide.                                                                                                                                                               |
| (e) the synthesis and breakdown of a disaccharide and polysaccharide by the formation and breakage of glycosidic bonds                                          | To include the disaccharides sucrose, lactose and maltose.                                                                                                                                                                                                                                                                              |
| (f) the structure of starch (amylose and amylopectin), glycogen and cellulose molecules                                                                         | HSW8                                                                                                                                                                                                                                                                                                                                    |
| (g) how the structures and properties of glucose, starch, glycogen and cellulose molecules relate to their functions in living organisms                        | HSW2, HSW8                                                                                                                                                                                                                                                                                                                              |

| Learning outcomes                                                                                                                     | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>(h)</b> the structure of a triglyceride and a phospholipid as examples of macromolecules                                           | To include the structure of saturated and unsaturated fatty acids.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>(i)</b> the synthesis and breakdown of triglycerides by the formation and breakage of ester bonds between fatty acids and glycerol | <p>To include hydrophobic and hydrophilic regions and energy content.</p> <p>Learners should be able to apply their knowledge and understanding in the context of prokaryotes and eukaryotes.</p> <p>HSW2, HSW8</p>                                                                                                                                                                                                                                                                                                                                   |
| <b>(j)</b> how the properties of triglyceride, phospholipid and cholesterol molecules relate to their functions in living organisms   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>(k)</b> the general structure of an amino acid                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>(l)</b> the synthesis and breakdown of dipeptides and polypeptides, by the formation and breakage of peptide bonds                 | <p>To include primary, secondary, tertiary and quaternary structure</p> <p><b>AND</b></p> <p>hydrogen bonding, hydrophobic and hydrophilic interactions, disulfide bonds and ionic bonds.</p> <p>HSW8</p>                                                                                                                                                                                                                                                                                                                                             |
| <b>(m)</b> the levels of protein structure                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>(n)</b> the structure and function of globular proteins including a conjugated protein                                             | <p>To include haemoglobin as an example of a conjugated protein (globular protein with a prosthetic group), a named enzyme and insulin.</p> <p><b>PAG10</b></p>                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>(o)</b> the properties and functions of fibrous proteins                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>(p)</b> the key inorganic ions that are involved in biological processes                                                           | <p>To include the following:</p> <p>calcium ions (<math>\text{Ca}^{2+}</math>), sodium ions (<math>\text{Na}^+</math>), potassium ions (<math>\text{K}^+</math>), hydrogen ions (<math>\text{H}^+</math>), ammonium ions (<math>\text{NH}_4^+</math>)</p> <p>nitrate (<math>\text{NO}_3^-</math>), hydrogencarbonate (<math>\text{HCO}_3^-</math>), chloride (<math>\text{Cl}^-</math>), phosphate (<math>\text{PO}_4^{3-}</math>), hydroxide (<math>\text{OH}^-</math>).</p> <p>Learners should be able to recognise the symbols for these ions.</p> |

| Learning outcomes                                                                                                                                                                                                                                                                                                | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                        |
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| <p><b>(q)</b> how to carry out and interpret the results of the following chemical tests:</p> <ul style="list-style-type: none"> <li>• biuret test for proteins</li> <li>• Benedict's test for reducing and non-reducing sugars</li> <li>• iodine test for starch</li> <li>• emulsion test for lipids</li> </ul> | <p><b>PAG9</b><br/>HSW3, HSW4, HSW5</p>                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>(r)</b> quantitative methods to determine the concentration of a chemical substance in a solution</p>                                                                                                                                                                                                      | <p>To include colorimetry.<br/><b>PAG5</b><br/>HSW3, HSW4, HSW5</p>                                                                                                                                                                                                                                                                                                                                        |
| <p><b>(s)</b> <b>(i)</b> the principles and uses of paper and thin layer chromatography to separate biological molecules / compounds</p> <p><b>(ii)</b> practical investigations to analyse biological solutions using paper or thin layer chromatography</p>                                                    | <p>To include calculation of <math>R_f</math> values.</p> $R_f = \frac{\text{distance moved by the solute}}{\text{distance moved by the solvent}}$ <p>Learners should be able to apply their knowledge and understanding in the context of separating proteins, carbohydrates, vitamins and nucleic acids.</p> <p><i>M0.1, M0.2, M1.1, M1.3, M2.2, M2.3, M2.4</i><br/><b>PAG6</b><br/>HSW2, HSW3, HSW4</p> |